

Unnamed_Unit

unit6 Expanding Knowledge on the Topic

PDF Documents

PDF Document 1: 1731845322854-Chapter 6_ unit6 Expanding Knowledge on the Topic.pdf

This PDF document is included in the unit materials.



UNIT 6 EXPANDING KNOWLEDGE ON THE TOPIC

The Fascinating Science of Fire Colors



WHAT MAKES FIRE DIFFERENT COLORS?

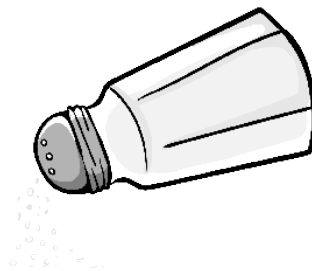
Have you ever wondered why fire isn't always the same color? Sometimes it's blue, sometimes it's red, and if you've seen fireworks, you know it can even be green or purple! Let's discover why this happens and what these colors can tell us.

WHY DOES FIRE CHANGE COLOR?

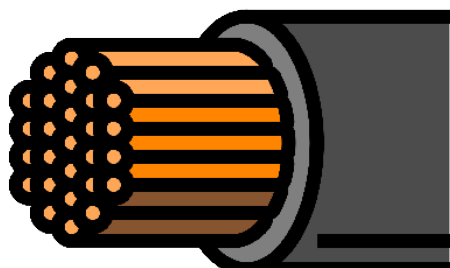
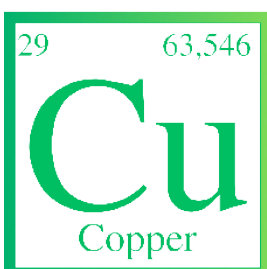
The color of a flame is primarily determined by two factors: the temperature of the fire and the chemicals that are burning within it. Higher temperatures typically result in blue or white flames, indicating that the fire is burning more efficiently and at a hotter temperature. On the other hand, cooler temperatures often produce red or yellow flames.

THE ROLE OF CHEMICALS

Different chemicals burn with different colors. For example:

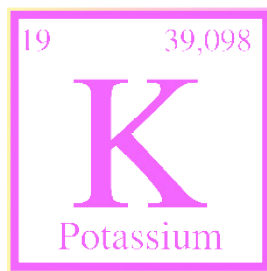


SODIUM burns with a bright yellow flame, which is why you see **intense yellow colors** in street lamps.



COPPER produces a green flame, often used in fireworks to create **dazzling green** bursts.

POTASSIUM burns with a purple flame, adding cool purple effects to firework shows.



SAFETY IN COLORS

Emergency responders also pay close attention to flame colors when identifying and handling hazardous materials.

For instance, a green flame can indicate the presence of barium, a toxic substance, necessitating specific safety measures.



TRY THIS AT HOME (WITH ADULT SUPERVISION!)

You can see different flame colors yourself by trying a safe experiment. If you burn a little bit of table salt in a fire (always with an adult to help!), you'll see the yellow flame because of the sodium in the salt.



WHAT WE LEARNED

Fire isn't just something that keeps us warm or cooks our food; it can be a rainbow of colors depending on what's burning and how hot it is. Each color tells a story about the fire, and by learning about these colors, we can understand more about science and safety.